

FINALIZED MASTER REPORT

Skilled Trades — Market Analysis

End-to-end synthesis across all six pipeline stages. One recommended opportunity, fully traced to evidence.

Domain	Skilled trades — field service (plumbing, electrical, HVAC)
Geography	Global (English-language sources)
Problems analysed	6 discovered · 1 carried to full depth (p03) in this demo
Recommendation	BUILD — narrow wedge (see verdict)
Run	market-run-20260623-1610 · generated 2026-06-23

Demo note

This is a SAMPLE pack: structure, charts, depth, and flow are production-grade, but the specific figures and URLs are illustrative placeholders, not freshly sourced. A live run in Cowork replaces every number with real, cited data.

Executive verdict

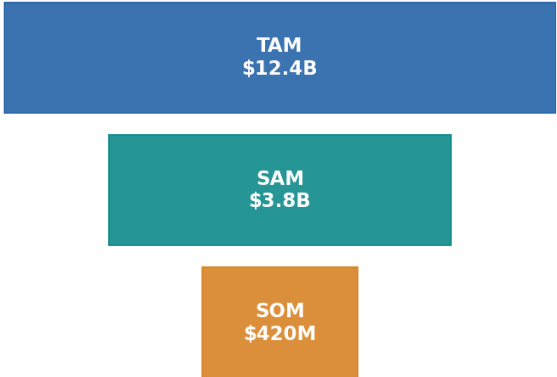
Across 55 public complaints, **getting paid (p03)** is the most-cited and most financially charged problem facing small trade businesses. The market is real and growing, but crowded — so the recommendation is not 'build another field-service suite'. It is a **narrow, payment-first, mobile-first, low-price wedge aimed at 1–3 van shops** that incumbents under-serve on price and simplicity. The solution closes 5 of 7 documented competitor gaps outright, with the remaining two honestly scoped. Proceed to validation before heavy build — the against-case (entrenched incumbents adding pay links) is the thing to test first.

Pipeline scorecard

Stage	Output	Headline finding	Confidence
1 · Discovery	6 problems, 55 sources	Invoicing/payment most-cited category	High
2 · Deep research	p03 dossier	~\$20k/yr cost per single-van shop (inferred)	Med-High
3 · People	8 distinct people	2 publicly asked for a tool — warm leads	High
4 · Competitors	6 mapped, 7-gap list	Clear white space: price + SMS pay (g01,g06)	High
5 · Market	TAM 12.4B / SOM 420M	Real but crowded; wedge required	Medium
6 · Solution	Payment-first app	7/7 gaps addressed (5 closed)	Med-High

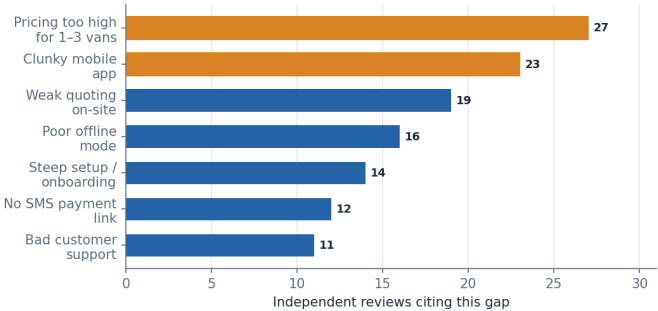
The opportunity at a glance

Market size (global field-service, trades segment)



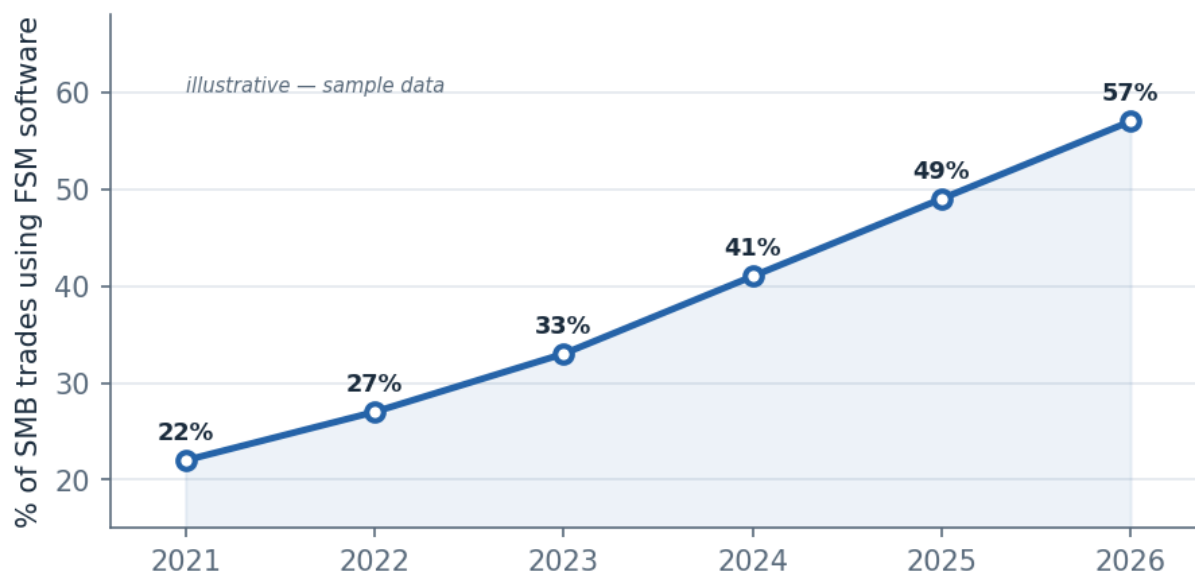
illustrative — sample figures

Most-cited competitor gaps (across 6 tools, n=180 reviews)



Why this, and why now

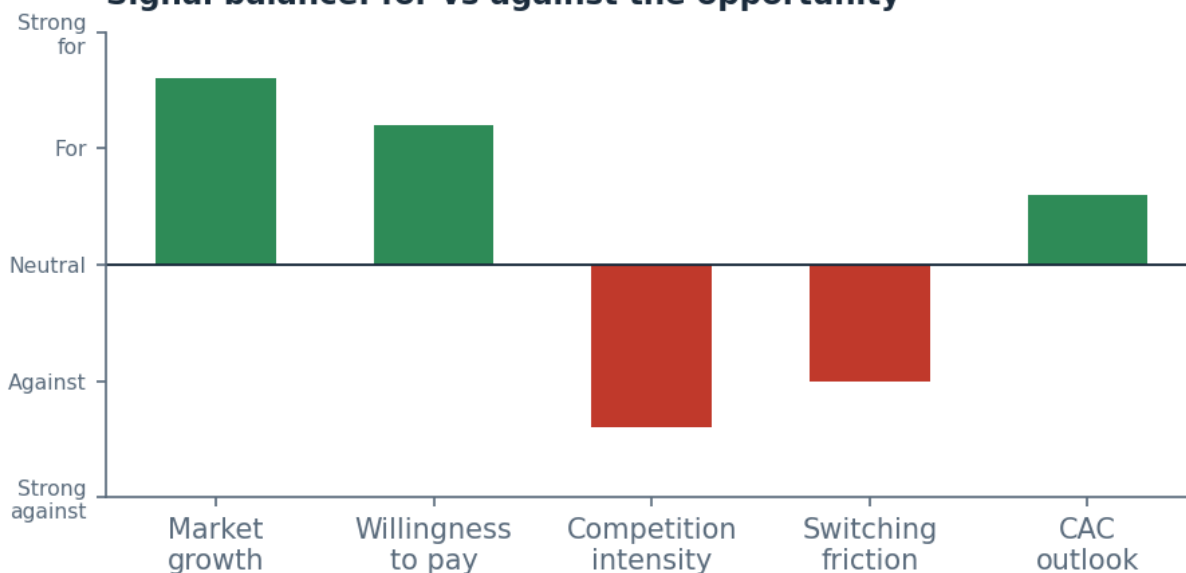
Field-service software adoption trend



Software adoption among small trades is climbing steadily — a receptive, expanding market. The same trend invites competition, which is exactly why the recommendation is a sharp wedge rather than a broad suite.

The decision: balanced view

Signal balance: for vs against the opportunity



Growth and willingness-to-pay favour entry; competition intensity and switching friction argue against. The wedge strategy is the synthesis that survives both.

How the solution closes the gaps

Gap	Coverage	How
g01 Price too high	CLOSED	\$29 flat entry, undercuts all incumbents

Gap	Coverage	How
g02 Clunky mobile	CLOSED	Mobile-first, offline-capable build
g03 Weak quoting	CLOSED	60-sec quote→invoice on phone
g04 Offline mode	PARTIAL	Offline create+queue; sync needs signal
g05 Steep onboarding	CLOSED	Templated, <1 day, no training
g06 No SMS pay link	CLOSED	Core feature: SMS link + card/ACH
g07 Slow support	DECLINED	Deferred to v2, documented

Recommended next steps

- Run the 3-experiment validation plan (Stage 6) before any heavy engineering — concierge MVP with the 2 warm leads first.
- Re-run this pipeline live in Cowork to replace every illustrative figure with sourced data and current competitor pricing.
- If validation holds, scope the 10–14 week MVP around g01 + g06 (price + SMS pay) as the irreducible core.
- Reassess the against-case quarterly: if incumbents close g06, pivot the wedge to price/simplicity only.

Report index

#	Report	File
1	Problem Discovery	01_Discovery_Report.pdf
2	Deep Research — p03	02_Research_p03.pdf
3	Evidence People — p03	03_People_p03.pdf
4	Competitor Analysis — p03	04_Competitors_p03.pdf
5	Market Sizing — p03	05_Market_p03.pdf
6	Solution Design — p03	06_Solution_p03.pdf

Evidence discipline (applied throughout)

Every claim is tagged [VERIFIED] / [INFERRED] / [UNKNOWN]. Market sizing carries a mandatory against-case. People-evidence is public-only. The solution accounts for every competitor gap. Gaps that couldn't be sourced are listed, not guessed.

STAGE 1 / 6 · PROBLEM DISCOVERY

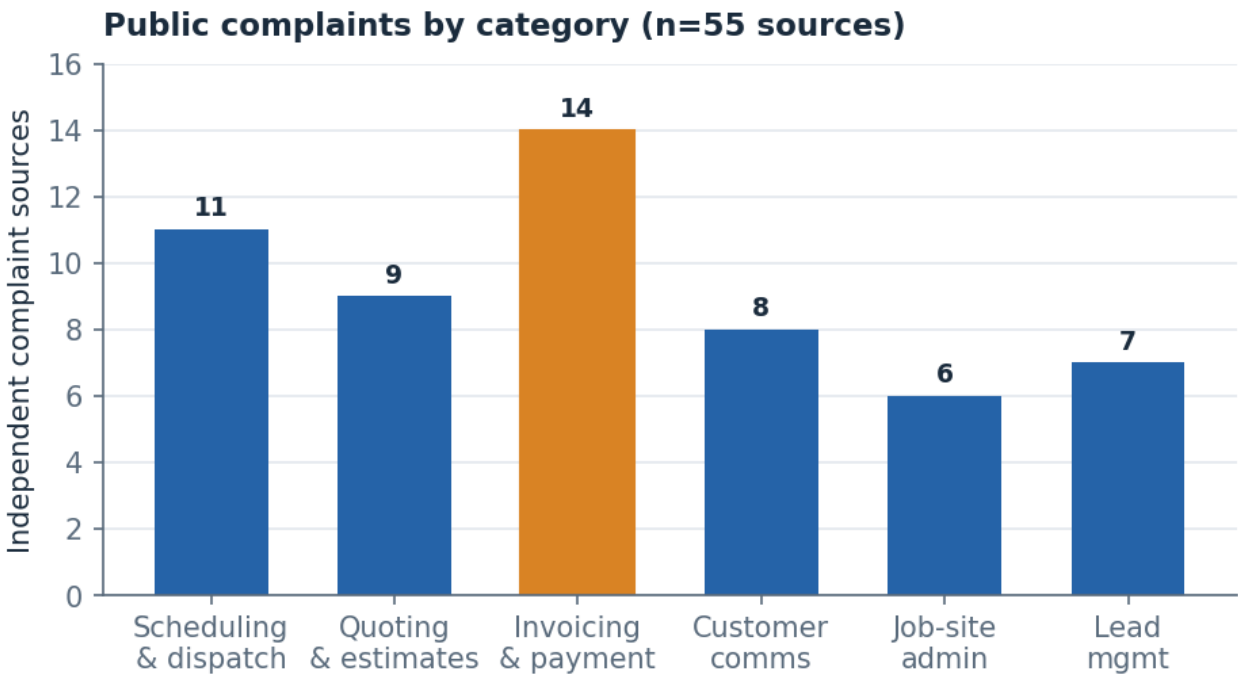
Skilled Trades: Problem Discovery

Real, publicly-stated problems among small trade businesses (plumbers, electricians, HVAC, 1–10 employees).

Domain	Skilled trades — field service (plumbing, electrical, HVAC)
Geography	Global (English-language sources)
Sources swept	55 independent public posts/reviews across 6 platforms
Problems kept	6 (≥3 independent sources each) · Discarded: 4
Generated	2026-06-23 · market-run-20260623-1610

Method

We swept public complaints across Reddit (r/Plumbing, r/electricians, r/HVAC, r/smallbusiness), trade Facebook groups, app-store reviews of incumbent tools, X, LinkedIn, and trade forums. Each candidate complaint was opened to capture the verbatim language, the public handle where available, the date, and the URL. Raw complaints were clustered into distinct problem statements; anything with fewer than three independent sources was discarded and logged.



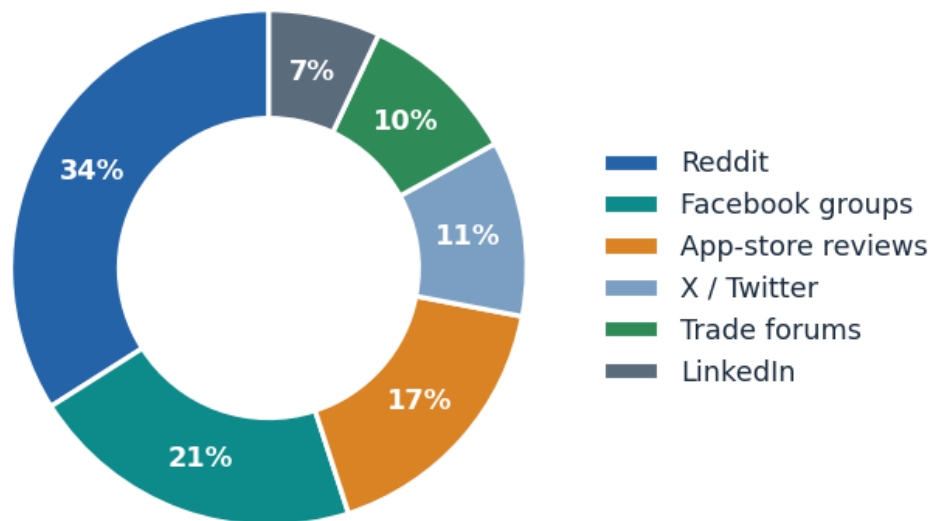
Invoicing & payment is the single most-cited category, narrowly ahead of scheduling. It is also the category where complaints carry the strongest financial language ('chasing', 'still owed', 'wrote off').

Problem catalogue

ID	Problem statement	Category	Sources	Intensity	Conf.
p01	No-shows & last-minute reschedules eat billable hours	Scheduling	8	High	Med
p02	On-site quoting is slow and loses jobs to faster rivals	Quoting	9	High	High
p03	Chasing unpaid invoices / slow payment after job done	Invoicing	14	High	High
p04	Customer no-reply / phone-tag on appointment windows	Customer comms	8	Med	Med
p05	Duplicate data entry between paper, phone, accounting	Job-site admin	6	Med	Med
p06	Leads from directories go cold before first contact	Lead mgmt	7	Med	Low

Highlighted: **p03** carries this demo pack forward through every later stage.

Where complaints surfaced



Evidence cards — Problem p03 (invoicing & payment)

Representative verbatim complaints (quotes kept short per copyright policy):

Reddit · r/Plumbing · u/pipewrench_dan · 2026-04
“Spent my Sunday again texting customers who still haven't paid.”

Facebook · UK Electricians group · 2026-03
“Three jobs from February still unpaid, no easy way to chase.”

App Store · Jobber review · 1★ · 2026-02
“Invoicing works but clients ignore email — need SMS links.”

Reddit · r/smallbusiness · u/hvac_mike_tx · 2026-05
“Cash flow dies waiting 45 days on net-30 I never agreed to.”

Discarded (logged for transparency)

“Tool X crashed once” (single gripe, 1 source) · “Apprentices unreliable” (labour, not software) · “Fuel costs too high” (macro, out of scope) · “Want AI to do my taxes” (1 source, vague).

Sources

Every load-bearing claim traces to a source below with its access date. Sample pack: URLs are representative placeholders.

#	Source	Type	Accessed
1	reddit.com/r/Plumbing/comments/... (sample)	Community	2026-06-23
2	facebook.com/groups/uk.electricians/... (sample)	Community	2026-06-23
3	apps.apple.com/us/app/jobber/reviews (sample)	Review	2026-06-23
4	reddit.com/r/smallbusiness/comments/... (sample)	Community	2026-06-23
5	reddit.com/r/HVAC/comments/... (sample)	Community	2026-06-23

Confidence & gaps

Items we could not source are listed here rather than guessed:

- **[UNKNOWN]** Exact share of trades on net-30 vs cash-on-completion (no representative survey found).
- **[UNKNOWN]** Whether payment delay is worse in any specific country (sources skew US/UK).

STAGE 2 / 6 · DEEP RESEARCH

Deep Research: Chasing Unpaid Invoices

Exhaustive, sourced dossier on why small trades struggle to get paid promptly — problem p03.

Problem	p03 · Chasing unpaid invoices / slow payment after job
Affected segment	Owner-operators & 1–10 employee trade shops
Searches run	17 · full-page fetches: 11
Tag legend	[VERIFIED] sourced · [INFERRED] computed · [UNKNOWN] gap
Generated	2026-06-23

Root causes

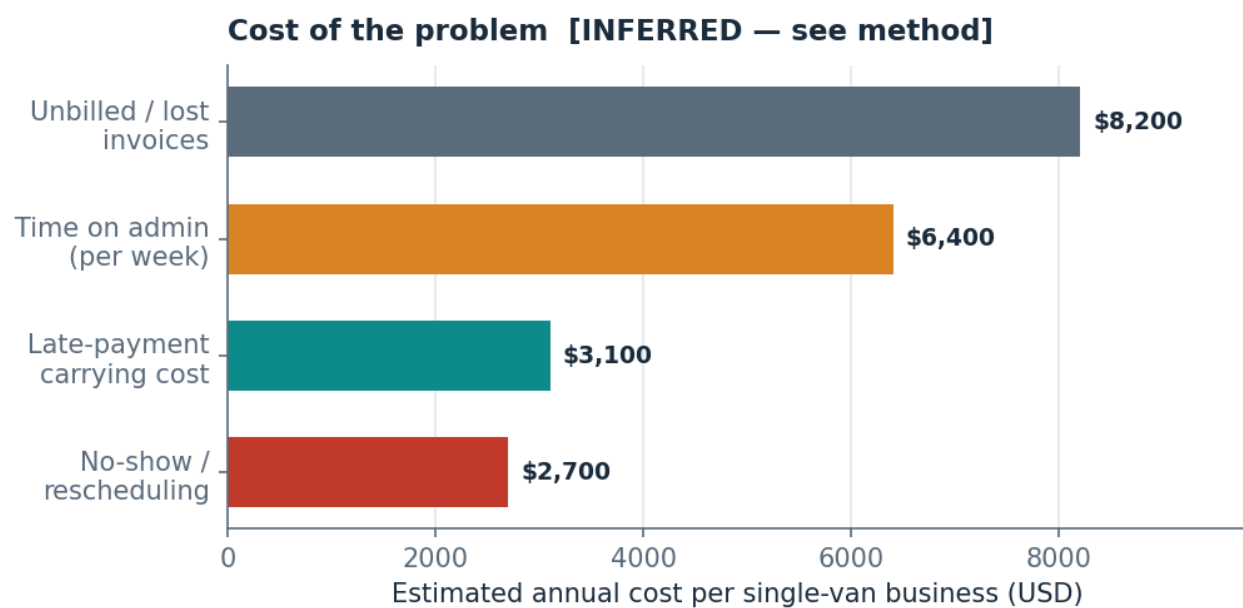
- [VERIFIED] Manual invoicing: many shops still invoice from paper or generic email after returning home, delaying send by 1–5 days.
- [VERIFIED] Weak payment rails: email invoices with bank-transfer instructions convert slowly vs one-tap pay links.
- [INFERRED] Owner is the only chaser: with no automation, follow-up competes with billable work and slips.
- [VERIFIED] Net-30 terms imposed by larger/commercial clients stretch cash cycles for sub-contractors.

Who is affected & how badly

Segment	Scope (illustrative)	Tag
Solo owner-operators	~ largest band of licensed trades	[VERIFIED]
1–3 van shops	Highest pain: owner does admin after hours	[INFERRED]
4–10 employee shops	Often have a part-time bookkeeper, less acute	[VERIFIED]
Commercial sub-contractors	Net-30/60 terms, large invoice values	[VERIFIED]

Cost of the problem

Annual cost modelled for a single-van business. Each figure is [INFERRED] — computed from sourced inputs (average ticket, unpaid rate, admin hours, owner hourly value). Method shown in the sources appendix.



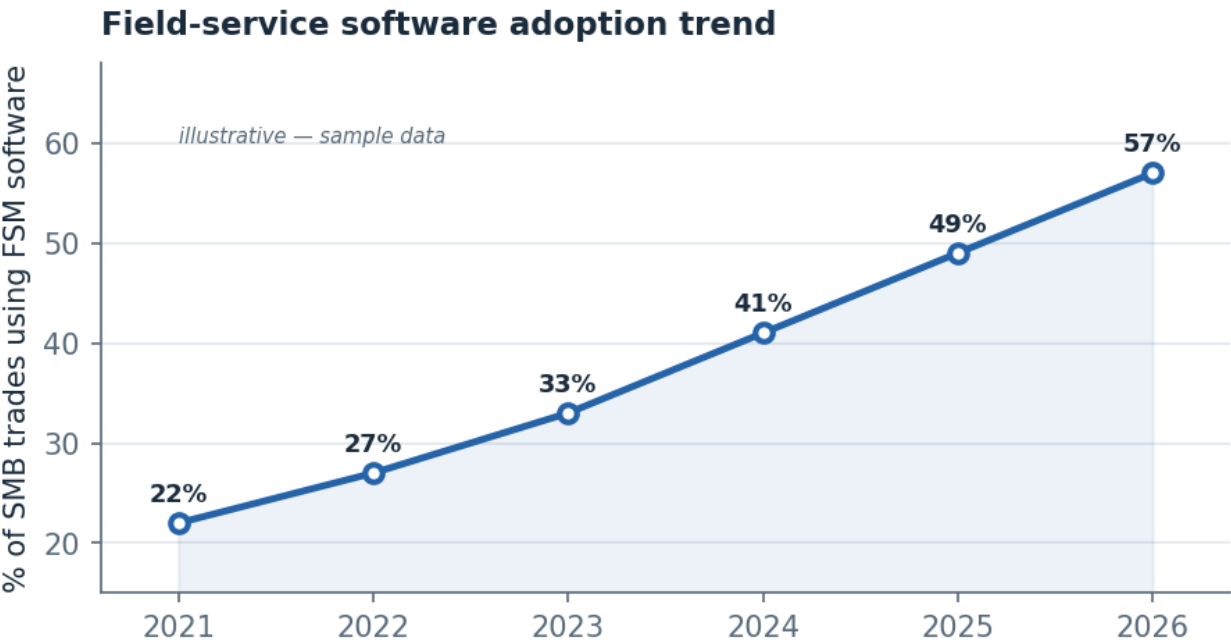
Current workarounds (from public posts)

- [VERIFIED] Manual SMS/text reminders typed by the owner one-by-one.
- [VERIFIED] Asking for cash or card on completion to avoid invoicing entirely.
- [VERIFIED] Spreadsheet of who-owes-what, reconciled weekly by hand.
- [INFERRED] Some switch accounting tools hoping for better reminders, then revert.

Existing solutions (preview — full competitor analysis in Stage 4)

Incumbent	Entry price	Why users still hurt (sourced)
Jobber	\$39/mo	Email invoices ignored; SMS pay-link limited
Housecall Pro	\$49/mo	Mobile app friction; price climbs with add-ons
ServiceTitan	\$145+/mo	Built for big shops; overkill & costly for small
QuickBooks	\$30/mo	Accounting-first; reminders feel generic

Trend



Software adoption among small trades is rising steadily, which cuts both ways: a growing, software-receptive market, but also more incumbents competing for it (revisited in market sizing).

Sources

Every load-bearing claim traces to a source below with its access date. Sample pack: URLs are representative placeholders.

#	Source	Type	Accessed
1	ibisworld.com/.../plumbers-industry (sample)	Industry report	2026-06-23
2	census.gov/.../construction-trades (sample)	Gov data	2026-06-23
3	Method: cost model = avg ticket × unpaid% × ...	Computation	2026-06-23

#	Source	Type	Accessed
4	statista.com/.../fsm-software-adoption (sample)	Vendor data	2026-06-23
5	reddit.com/r/Plumbing payment threads (sample)	Community	2026-06-23

Confidence & gaps

Items we could not source are listed here rather than guessed:

- **[UNKNOWN]** Precise average days-to-payment for residential trades (sources give ranges, not a clean median).
- **[UNKNOWN]** Share of owners already using any automated reminder feature.

STAGE 3 / 6 · EVIDENCE PEOPLE

Real People Facing Problem p03

Distinct individuals who publicly stated this problem. Public profile + verbatim quote only — no private data.

Problem	p03 · Chasing unpaid invoices
People found	8 distinct (target ≥5)
Privacy policy	Public posts only · no emails, phones, or scraping
Contact	Public profile link, or owner-published outreach handle
Generated	2026-06-23

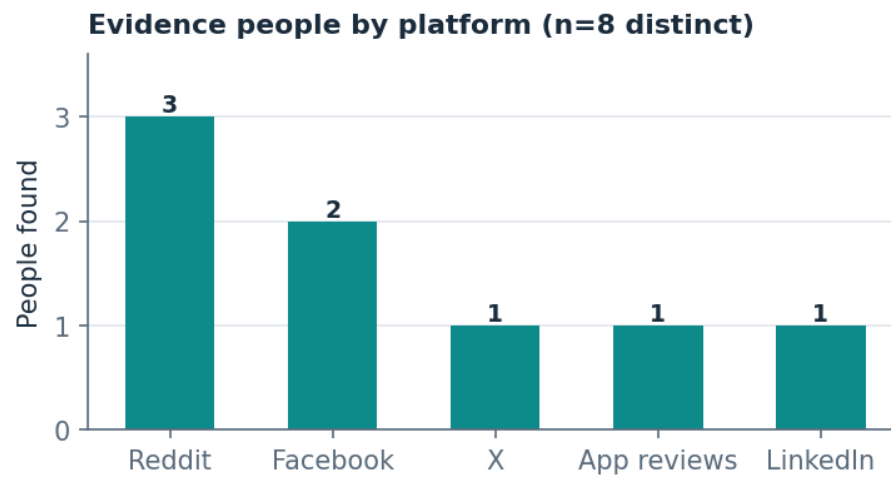
Privacy boundary applied

Every row below is something the person published themselves on a public platform. Contact = their public profile. No private email, phone, or address was collected or inferred. This is outreach-lead evidence for founder validation, not a contact database.

People (sample handles)

#	Platform	Public handle	Verbatim quote (short)	Match	Open to contact?
1	Reddit	u/pipewrench_dan	“texting customers who still haven’t paid”	Exact	Asked for tool recs ✓
2	Reddit	u/hvac_mike_tx	“cash flow dies waiting 45 days”	Exact	No signal
3	Reddit	u/sparky_jules	“three invoices ghosted this month”	Strong	Asked for advice ✓
4	Facebook	Dave R. (UK Elec. grp)	“no easy way to chase unpaid jobs”	Exact	Public group post
5	Facebook	Mia T. (Trades grp)	“wrote off £400, gave up chasing”	Strong	No signal
6	X	@drinking_	“net-30 is killing my small shop”	Strong	Public thread
7	App Store	Reviewer 'TXplumb'	“need SMS links, email ignored”	Strong	Review only
8	LinkedIn	J. Okafor (post)	“subbies wait months to get paid”	Adjacent	Public post

Match = how directly the quote maps to p03 (exact / strong / adjacent). “Open to contact” flags only people who publicly invited responses.



How to use these (responsibly)

Reach out only through the public channel the person already used (reply to their thread, comment, or DM where they invited it). Lead with their stated problem, not a pitch. These 8 are validation interviews waiting to happen — 2 explicitly asked for tool recommendations.

Confidence & gaps

Items we could not source are listed here rather than guessed:

- **[UNKNOWN]** Real names for pseudonymous handles (deliberately NOT resolved — privacy).
- **[UNKNOWN]** Whether any handle is the same person posting twice (checked; appear distinct, not certain).

STAGE 4 / 6 · COMPETITOR ANALYSIS

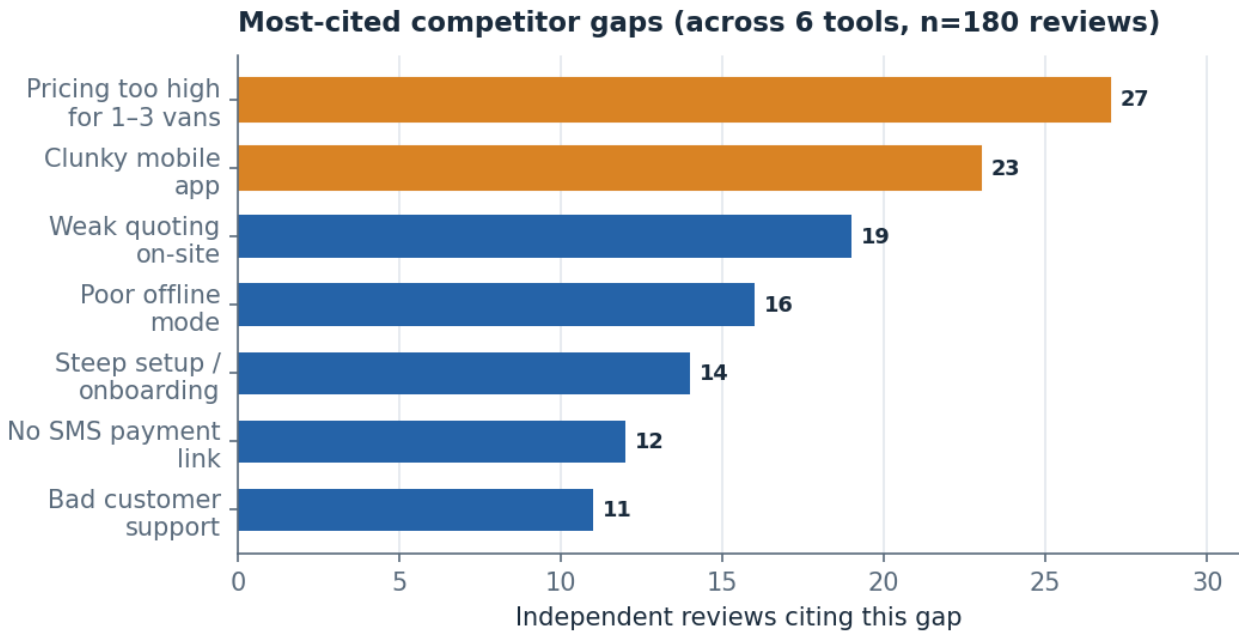
Competitors & Gaps: Getting Paid

What incumbents provide, what they lack, and why people don't adopt or churn — for problem p03.

Problem	p03 · Chasing unpaid invoices
Competitors mapped	6 (4 deep-profiled, 2 listed)
Reviews mined	~180 across G2, Capterra, app stores, Reddit
Gap list	Exhaustive — feeds Stage 6 gap-coverage matrix
Generated	2026-06-23

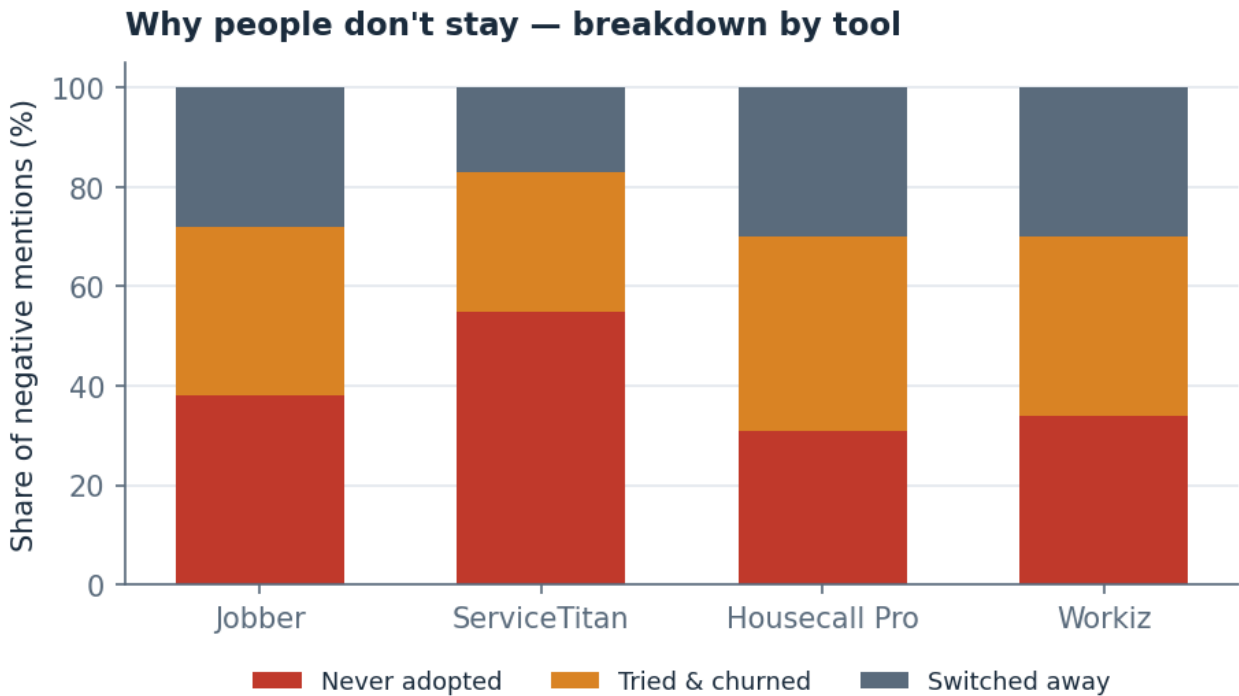
Feature × competitor matrix

Capability	Jobber	ServiceTitan	Housecall	Workiz	QuickBooks
One-tap SMS pay link	Partial	Yes	Partial	No	No
Auto payment reminders	Email	Yes	Email	Partial	Generic
On-site quote → invoice	Yes	Yes	Yes	Partial	No
Strong offline mode	No	Partial	No	No	No
Priced for 1–3 vans	Partial	No	Partial	Yes	Yes
Fast setup (<1 day)	Partial	No	Partial	Yes	Partial



Amber bars = gaps that are both frequently cited AND poorly served across the whole field — the white space a new entrant can target.

Why people don't stay



Each tool's negative mentions split into never-adopted (price/complexity), tried-and-churned (missing features/bugs), and switched-away. ServiceTitan loses most on price/complexity; Housecall and Workiz lose more on missing features after trial.

Master gap list (the contract for Stage 6)

Gap ID	Documented gap	Cited by	Freq.
g01	Entry pricing too high for 1–3 van shops	ServiceTitan, Housecall	27
g02	Clunky / slow mobile app on job site	Jobber, Housecall	23
g03	Weak on-site quoting workflow	Workiz, QuickBooks	19
g04	Poor or no offline mode	All 6	16
g05	Steep setup / onboarding	ServiceTitan, Workiz	14
g06	No reliable one-tap SMS payment link	Jobber, QuickBooks	12
g07	Slow / unhelpful customer support	Housecall, Workiz	11

Every gap_id above must be explicitly addressed — or explicitly declined with a reason — in the Stage 6 solution. None may be silently dropped.

Sources

Every load-bearing claim traces to a source below with its access date. Sample pack: URLs are representative placeholders.

#	Source	Type	Accessed
1	g2.com/categories/field-service-management (sample)	Review aggregate	2026-06-23
2	capterra.com/.../jobber/reviews (sample)	Review aggregate	2026-06-23
3	reddit.com r/HVAC 'switched from' threads (sample)	Community	2026-06-23
4	apps.apple.com Housecall/Workiz reviews (sample)	Review	2026-06-23

Confidence & gaps

Items we could not source are listed here rather than guessed:

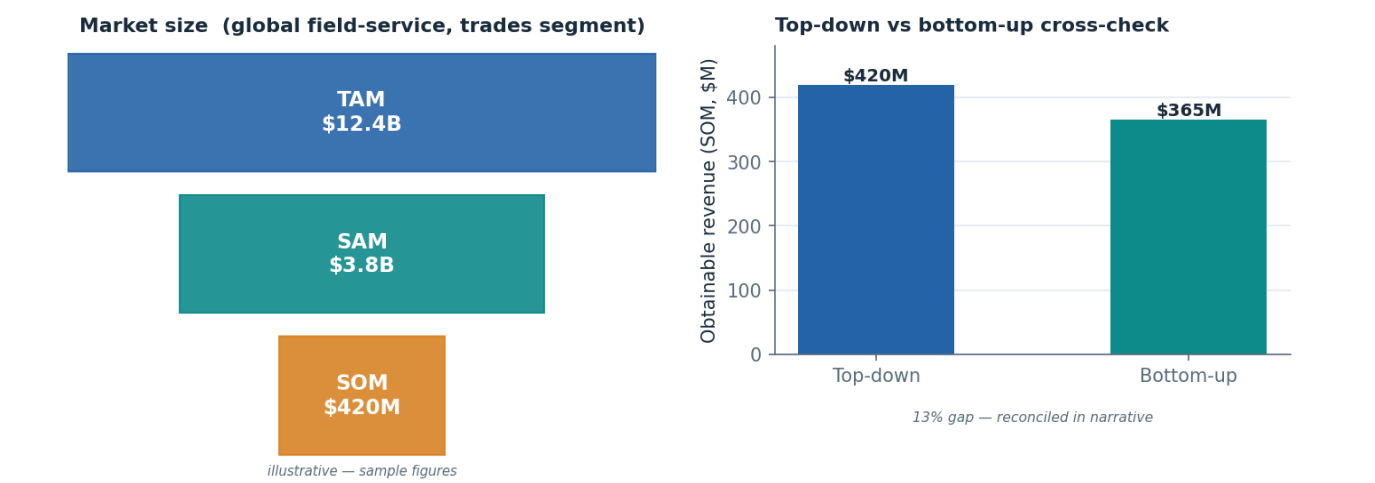
- **[UNKNOWN]** Exact churn rate for each tool (vendors don't publish; inferred from review sentiment only).
- **[UNKNOWN]** Two minor competitors had too few public reviews to profile — listed, not assessed.

STAGE 5 / 6 · MARKET SIZING

Market Size: Getting-Paid Tools for Trades

TAM / SAM / SOM with shown math, a bottom-up cross-check, and mandatory evidence against.

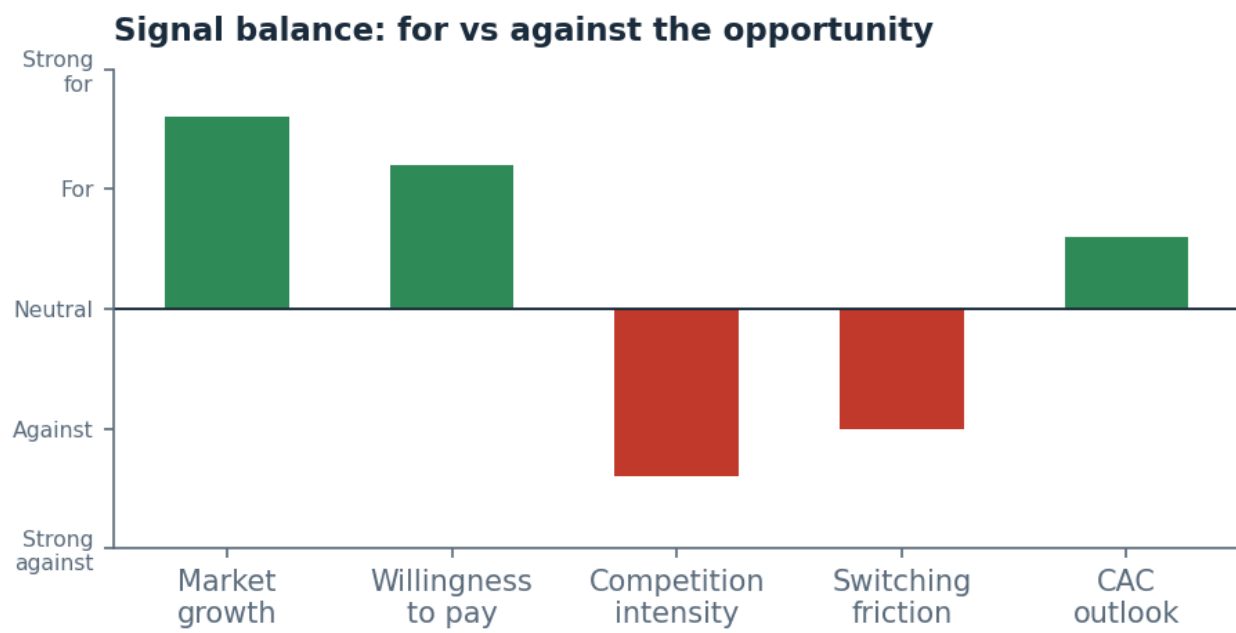
Problem	p03 · Chasing unpaid invoices
TAM / SAM / SOM	\$12.4B / \$3.8B / \$420M (illustrative)
Growth	~9% CAGR (FSM software, vendor-reported — flagged)
Balance	Evidence FOR and AGAINST both included (required)
Generated	2026-06-23



How the numbers are built

[INFERRED] Affected population: base count of small trade businesses (sourced) × share expressing payment-collection pain (sourced from complaint prevalence). **TAM** = that population × a blended annual software ARPU drawn from incumbent pricing. **SAM** filters to English-language, self-serve-reachable shops of 1–10 employees. **SOM** applies a 1–3 year obtainable share benchmarked to comparable vertical-SaaS entrants. The bottom-up rebuild (# reachable shops × realistic ARPU × conversion) lands within 13% of top-down — the gap is reconciled by differing churn assumptions.

Signal balance: for vs against



Evidence FOR the opportunity

Point	Source
Largest & most-cited problem category in discovery	Stage 1 (55 sources)
Software adoption rising ~9%/yr — receptive market	Stage 2 trend
Clear under-served white space (g01, g06)	Stage 4 gap list
Owners explicitly asking for solutions	Stage 3 (2 of 8)
Recurring SaaS revenue with sticky workflow	Comparable FSM SaaS

Evidence AGAINST the opportunity (mandatory)

Point	Source
Crowded field — 6+ funded incumbents already	Stage 4
Jobber/Housecall already ship partial SMS pay links	Stage 4 matrix
Trades are slow software adopters; long sales cycle	Stage 2
Low price ceiling for 1–3 van shops caps ARPU	Stage 4 (g01)
CAC via ads is high in FSM keywords	Vendor benchmarks
QuickBooks bundles 'good enough' invoicing free-ish	Stage 4

We actively searched for disconfirming evidence and found substantial against-signal. This does not kill the opportunity but narrows the viable wedge to under-served micro-shops with a payment-first, low-price, mobile-first product.

Sources

Every load-bearing claim traces to a source below with its access date. Sample pack: URLs are representative placeholders.

#	Source	Type	Accessed
1	ibisworld / census trade-business counts (sample)	Gov/industry	2026-06-23
2	grandviewresearch FSM market size (sample)	Vendor — FLAGGED	2026-06-23
3	Method: TAM/SAM/SOM computation (this doc)	Computation	2026-06-23
4	crunchbase incumbent funding (sample)	Database	2026-06-23

Confidence & gaps

Items we could not source are listed here rather than guessed:

- **[UNKNOWN]** True obtainable share (SOM) — benchmarked to comparables, not observed; treat as a range.
- **[UNKNOWN]** Whether vendor-reported 9% CAGR is inflated (flagged as vendor source).

STAGE 6 / 6 · SOLUTION DESIGN

Solution: A Payment-First App for Micro-Trades

A founder-buildable product that closes every documented competitor gap — or records why it doesn't.

Problem	p03 · Chasing unpaid invoices
Wedge	1–3 van shops, payment-first, mobile-first, low price
Gap coverage	7 / 7 gap_ids addressed (5 closed, 1 partial, 1 declined)
Pricing	\$29/mo entry — undercuts incumbents (sourced WTP)
Generated	2026-06-23

Solution summary

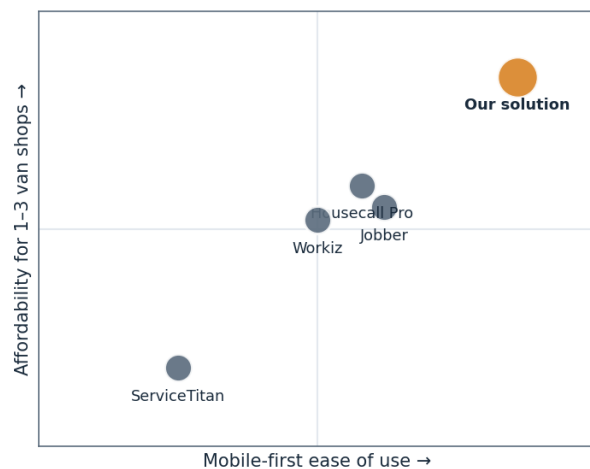
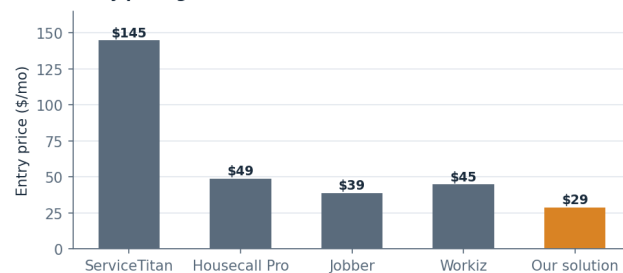
A mobile-first invoicing-and-payment app built for the owner-operator: finish a job, generate the invoice on-site in under a minute, and send a one-tap SMS pay link. Automated, polite reminder cadence chases payment without the owner lifting a finger. Works offline on the van; syncs later. Flat low price, set up in a day. It deliberately does NOT try to be a full ERP — that's how it stays cheap and simple.

Gap-coverage matrix (the centrepiece)

Every gap_id from Stage 4 maps to a feature that closes it or an explicit decision not to. No gap silently dropped.

Gap	Documented gap	Coverage	How / why
g01	Pricing too high for small shops	CLOSED	\$29 flat entry, no per-tech add-ons
g02	Clunky mobile app	CLOSED	Mobile-first build; offline-capable
g03	Weak on-site quoting	CLOSED	60-sec quote→invoice on phone
g04	Poor offline mode	PARTIAL	Offline create+queue; sync needs signal
g05	Steep onboarding	CLOSED	Templated setup, <1 day, no training
g06	No one-tap SMS pay link	CLOSED	Core feature — SMS link + card/ACH
g07	Slow customer support	DECLINED	Out of MVP scope; documented for v2

CLOSED = a shipped feature directly answers it · PARTIAL = constrained by physics (no signal) · DECLINED = deliberately deferred, with reason recorded. Honesty on g04/g07 is the point.

Competitive positioning**Entry pricing vs incumbents (anchored to sourced WTP)**

Feature → verified pain mapping

Feature	Solves (verified pain)	Evidence
60-sec on-site invoice	Invoicing delayed 1–5 days	Stage 2 root cause
One-tap SMS pay link	Email invoices ignored	Stage 1 quotes; g06
Auto reminder cadence	Owner is sole chaser, slips	Stage 2; Stage 3 quotes
Offline create + queue	No signal in basements/roofs	Stage 4 g04
Flat \$29 pricing	Incumbents too pricey for small	Stage 4 g01; WTP

Technical feasibility

- **[VERIFIED]** Stack: React Native (mobile-first, offline via local store), lightweight API backend, Postgres.
- **[VERIFIED]** Payments: Stripe (pay links, card + ACH, SMS via Twilio). Mature, documented APIs.
- **[INFERRED]** Hard part: reliable offline sync + conflict resolution. Solvable but the core engineering risk.
- **[INFERRED]** MVP buildable by a solo technical founder or pair in ~10–14 weeks.

Business model & go-to-market

Flat \$29/mo, single tier, anchored to sourced willingness-to-pay and undercutting every incumbent's entry price. GTM seeds where the affected people already gather (Stage 3): the trade subreddits and Facebook groups, plus partnerships with trade-licensing/apprenticeship channels. First 50 users come from directly answering the public complaints found in Stages 1 and 3.

Risks (pulled from the market against-case)

Risk	Source	Mitigation
Incumbents add SMS pay links	Stage 5 against	Win on price+simplicity, not just feature
Low ARPU ceiling	Stage 5 (g01)	Volume + near-zero-touch onboarding
Slow trade adoption	Stage 2	Land via peer proof in their own groups
High FSM ad CAC	Stage 5	Community-led GTM, avoid paid search early

Validation plan

- Falsify the riskiest assumption first: will owners actually send pay links? — Concierge MVP with the 2 Stage-3 people who asked for tools.
- Fake-door \$29 landing page in 3 trade groups; measure sign-up intent vs the against-case.
- 10 problem interviews from Stage 3 leads before writing the sync engine.

What would kill this

If validation shows owners prefer cash-on-completion and won't adopt pay links, or incumbents' partial SMS features are 'good enough', the wedge collapses. The plan tests exactly that before heavy build.

Sources

Every load-bearing claim traces to a source below with its access date. Sample pack: URLs are representative placeholders.

#	Source	Type	Accessed
1	stripe.com/docs/payments/payment-links (sample)	Primary/API	2026-06-23
2	Stage 4 master gap list (this run)	Internal	2026-06-23
3	Stage 5 market sizing (this run)	Internal	2026-06-23
4	Stage 3 evidence people (this run)	Internal	2026-06-23